

WASP-52b

R-1082

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_151112 · created 2026-07-30T15:43:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457338.7361 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.062
Transit depth (Rp/Rs)^2	2.59 +/- 2.01 [%]
Orbital Inclination (inc)	81.65 +/- 2.8
Airmass coefficient 1 (a1)	0.97 +/- 0.62
Airmass coefficient 2 (a2)	0.05 +/- 0.33
Scatter in the residuals of the lightcurve fit is	69.11 %
Best Comparison Star	#1 - [317, 228]
Optimal Aperture	1.34
Optimal Annulus	3.0
Transit Duration (day)	0.034 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.161 ± 0.062 vs 0.1646 (deviation 0.05σ)
Duration fit vs published	0.034 d vs 0.0758 d (deviation 1.15σ)
Mid-time O–C	-3.5 min
Depth SNR	2.2
Residual scatter	69.11 %

Light curve

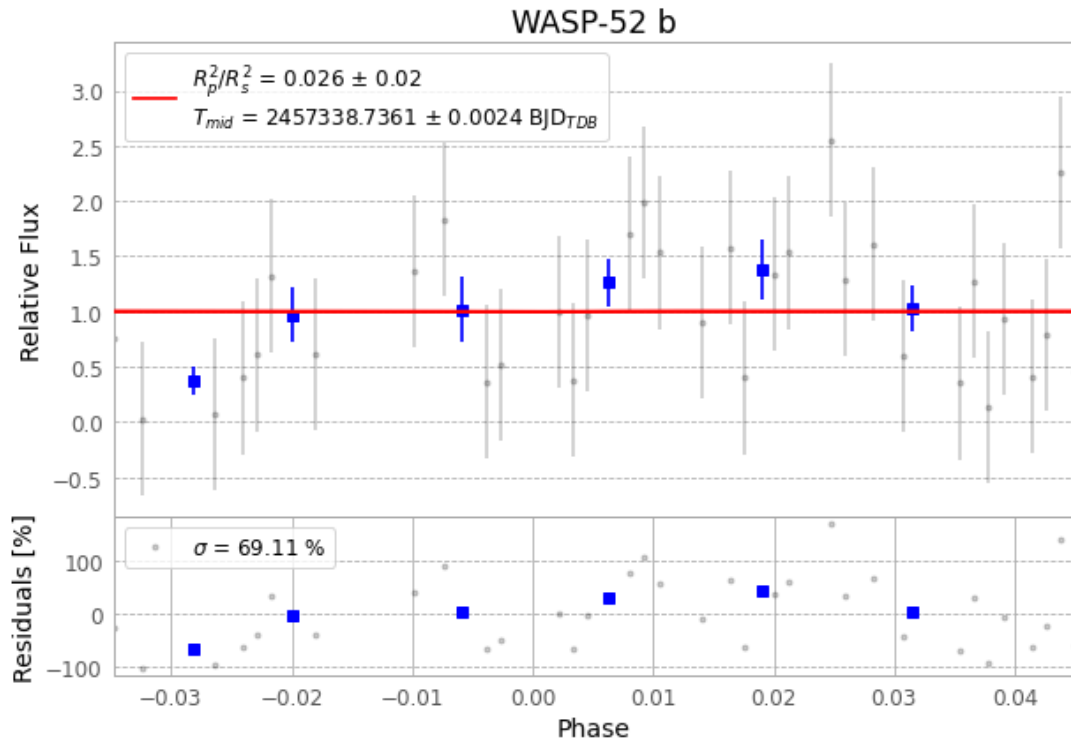


Figure 1 — FinalLightCurve_WASP-52 b_2015-11-11.png (SHA-256 a05b46ed5e885b10...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [526, 401], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 cca510a659ec5226...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), WASP-52151112040012.FITS → WASP-52151112073612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-151112.log (2f1ae1c193cd...), FinalLightCurve_WASP-52 b_2015-11-11.png (a05b46ed5e88...), FinalLightCurve_WASP-52 b_2015-11-11.csv (3e4ffe5fb6d5...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 15:43Z.